

Web scraping from Wikipedia



in the era of LLMs

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The presentation is based on a personal project, done in my free time using my own resources. I am not representing my employer.

Wikipedia: Five Pillars



Wikipedia is an encyclopedia



Respect and civility between editors



Written from a neutral point of view



No firm rules



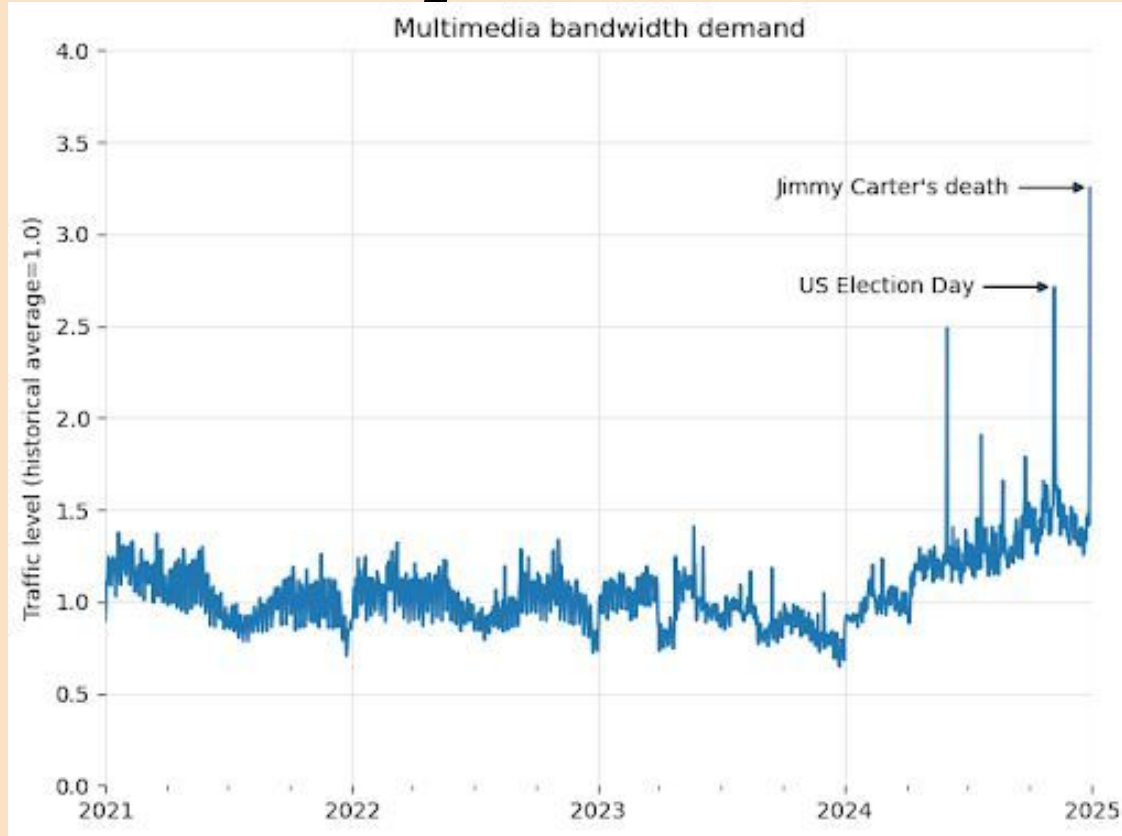
Anyone can use, edit and distribute



AI scrapers impact on Wikipedia

Maciej Artur Nadzikiewicz
Board @ Wikipedia
LinkedIn

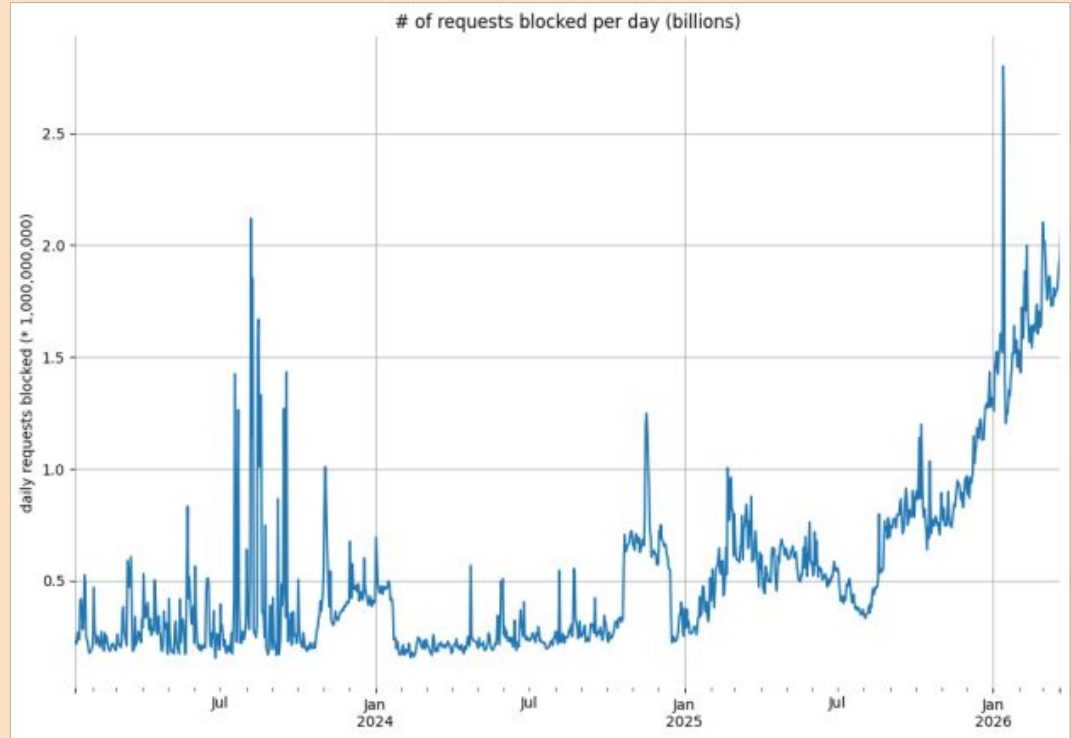
Costs of Wikipedia infrastructure have grown by 50% since January 2024 due to bots and AI scrapers. 65% of our most expensive traffic comes from bots, a disproportionate amount given that bot pageviews account for around 1/3 of the total.



Wikipedia blocking crawler requests

"we're currently blocking or throttling about 25% of all automated requests that are coming from crawlers that don't adhere to our policies (up to billions of requests per day)."

<https://diff.wikimedia.org/2026/03/26/quo-vadis-crawlers-progress-and-whats-next-on-safeguarding-our-infrastructure/>



Wikimedia Projects

Projects of the Wikimedia Foundation



Wikipedia

The Free Encyclopedia



Commons

Free media collection



Wikibooks

Free textbooks



Wikidata

Free knowledge base



Wikifunctions

Free function library



Wikinews

Free news source



Wikiquote

Free quote
compendium



Wikisource

Free content library



Wikispecies

Free species directory



Wikiversity

Free learning resources



Wikivoyage

Free travel guide



Wiktionary

Free dictionary



MediaWiki

Free & open wiki
software



Meta-Wiki

Community
coordination &
documentation

Licensing considerations



Wikipedia Text

- CC BY-SA 4.0
- GPLv3 compatible with it

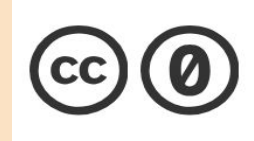


Media: Images, video, audio

- public domain,
- non commercial use, no derivative works
- other

Wikidata structured data from the main, Property, Lexeme, and EntitySchema namespaces

- CC0



A historical term in british law

common. 1. (Engl.) Species of PROFIT À PRENDRE; a right enjoyed in another's land in common with him or others, e.g. common of pasture (the right to graze beasts on the land of another), of piscary (to fish), of estovers (to cut wood), of turbary (to dig turf), in the soil (to dig minerals, etc.)

2. Land over which rights of common (or, in Sc., COMMONTY) are enjoyed;

The Reader's Digest Great Encyclopaedic Dictionary 3rd edition, 1978

The tragedy of the Commons

Garrett Hardin

using the word "tragedy" as the philosopher Whitehead used it (7): "The essence of dramatic tragedy is not unhappiness. It resides in the solemnity of the remorseless working of things."

The tragedy of the commons develops in this way. Picture a pasture open to all. It is to be expected that each herdsman will try to keep as many cattle as possible on the commons.

The rational herdsman concludes that the only sensible course for him to pursue is to add another animal to his herd. And another; and another.... But this is the conclusion reached by each and every rational herdsman sharing a commons. Therein is the tragedy.

Ancient tragedy with a modern twist

- Behavioural Science shows that humans often don't act as "rational and self-interested"
 - Volunteer work of Wikipedians: editors and administrators
- Two concepts from Stefaan G. Verhulst
- The Weaponisation of openness (Oct 2025)
 - *open resources have been mined at planetary scale for AI training. This process has largely taken place without meaningful consent, compensation, or even attribution. The result is a modern-day twist on the tragedy of the commons: The more open the digital ecosystem became, the more it was extracted.*
- The Data Winter (Jan 2024):
 - *we may be entering a prolonged data winter [...] in which data assets that could be leveraged for the common good are instead frozen and immobilized.*

Extraction/mining a better analogy?



-In the old days

```
%pip install lxml
```

```
import pandas as pd
```

```
wiki_tables=pd.read_html('https://en.wikipedia.org/wiki/Northern_Sea_Route')
```

```
wiki_tables[0]
```

Year	Total	Russia	Singapore	Finland	Norway
2007	2	2.0	NaN	NaN	NaN
2008	3	3.0	NaN	NaN	NaN
2009	5	5.0	NaN	NaN	NaN

Current policy

```
import pandas as pd
import wikipedia
import wget
import io

%xmode Minimal
wiki_tables=pd.read_html('https://en.wikipedia.org/wiki/Northern_Sea_Route')
... Exception reporting mode: Minimal
... HTTPError: HTTP Error 403: Forbidden
wiki_page=wikipedia.page('Northern_Sea_Route',auto_suggest = False).html()
wiki_tables2=pd.read_html(io.StringIO(wiki_page))
Wiki_tables2[0]
```

Year	Total	Russia	Singapore	Finland	Norway	Germany	Spain
2007	2	2.0	NaN	NaN	NaN	NaN	NaN
2008	3	3.0	NaN	NaN	NaN	NaN	NaN
2009	5	5.0	NaN	NaN	NaN	NaN	NaN

Wikipedia Library

- `wikipedia.page('Title')`
 - `search`
- `wikipedia.Wikipage.__init__()`
- `_wiki_request()`

Func search

Class `WikipediaPage`

Func `html`

Func `content`

Func `revision_id`

Func `parent_id`

Func `summary`

Func `images`

Func `coordinates`

Func `references`

Func `add_protocol`

Func `links`

Func `categories`

Func `sections`

Func `section`

Func `_wiki_request`

Coordinates

```
air_base=wikipedia.page("Olenya_air_base").coordinates
air_base
...(Decimal('68.15166666999999733889126218855381011962890625'),
  Decimal('33.465000000000003410605131648480892181396484375'))
```

Images

```
wiki_DOP=wikipedia.page("Dilution_of_precision_(navigation)")
wiki_DOP.images[0]
...'https://upload.wikimedia.org/wikipedia/commons/c/c3/Bad\_gdop.png'

wget.download(wiki_DOP.images[0])
... HTTPError: HTTP Error 403: Forbidden

!wget https://upload.wikimedia.org/wikipedia/commons/c/c3/Bad\_gdop.png
...--2026-06-08 19:43:39-- https://upload.wikimedia.org/wikipedia/commons/c/c3/Bad\_gdop.png
...Resolving upload.wikimedia.org (upload.wikimedia.org)... 185.15.59.240, 2a02:ec80:300:ed1a::2:b
...Connecting to upload.wikimedia.org (upload.wikimedia.org)|185.15.59.240|:443... connected.
...HTTP request sent, awaiting response... 200 OK
...Length: 868595 (848K) [image/png]
...Saving to: 'Bad_gdop.png'

...Bad_gdop.png          100%[=====>] 848.24K  --.-KB/s    in 0.05s

...2026-06-08 19:43:39 (17.0 MB/s) - 'Bad_gdop.png' saved [868595/868595]
```

-Dirty data because of footnotes

```
problem_table.iloc[550:]
```

	IATA	ICAO	Airport name	Location served	Time	DST
550	AZL	SWTU[4]	Fazenda Tucunaré Airport[1]	Sapezal, Mato Grosso, Brazil	UTC-04:00	NaN
551	AZN	UTFA	Andizhan Airport	Andizhan, Uzbekistan	UTC+05:00	NaN
552	AZO	KAZO	Kalamazoo/Battle Creek International Airport	Kalamazoo, Michigan, United States	UTC-05:00	Mar-Nov
553	AZP	MMJC	Jorge Jiménez Cantú National Airport	Ciudad López Mateos, State of Mexico, Mexico	UTC-06:00	Apr-Oct
554	AZR	DAUA	Touat-Cheikh Sidi Mohamed Belkebir Airport	Adrar, Algeria	UTC+01:00	NaN
555	AZS	MDCY	Samaná El Catey International Airport	El Catey, Dominican Republic	UTC-04:00	NaN
556	AZT	NaN	Zapatoca Airport	Zapatoca, Colombia	UTC-05:00	NaN
557	AZZ	FNAM	Ambriz Airport	Ambriz, Angola	UTC+01:00	NaN

Non breaking space

```
problem_table["Airport name"]
...KeyError: 'Airport name'

problem_table.iloc[1].to_dict()

{('IATA', '-AA-'): 'AAB',
 ('ICAO', '-AA-'): 'YARY',
 ('Airport\\xa0name', '-AA-'): 'Arrabury Airport',
 ('Location\\xa0served', '-AA-'): 'Arrabury, Queensland, Australia',
 ('Time', '-AA-'): 'UTC+10:00',
 ('DST', '-AA-'): nan}
```

#in html it looks like

```
<th style="text-align:left;">Airport&#160;name
</th>
<th style="text-align:left;">Location&#160;served
</th>
```

WikiData

```
from SPARQLWrapper import SPARQLWrapper, JSON
import pandas as pd

sparql = SPARQLWrapper("https://query.wikidata.org/sparql")
sparql.setQuery("""
SELECT ?item ?itemLabel
WHERE
{
    ?item wdt:P31 wd:Q146 .
    SERVICE wikibase:label { bd:serviceParam wikibase:language "en" }
}
""")

sparql.setReturnFormat(JSON)
results = sparql.query().convert()
results_df = pd.json_normalize(results['results']['bindings'])
results_df[['item.value', 'itemLabel.value']].head()
```

	item.value	itemLabel.value
0	http://www.wikidata.org/entity/Q378619	CC
1	http://www.wikidata.org/entity/Q498787	Muezza
2	http://www.wikidata.org/entity/Q677525	Orangey

Snapshot API/Kaggle



Data Fields

The data fields are the same among all, noteworthy included fields:

name - title of the article.

identifier - ID of the article.

url - URL of the article.

version: metadata related to the latest specific revision of the article

version.editor - editor-specific signals that can help contextualize the revision

version.scores - returns assessments by ML models on the likelihood of a revision being reverted.

main entity - Wikidata QID the article is related to.

abstract - lead section, summarizing what the article is about.

description - one-sentence description of the article for quick reference.

image - main image representing the article's subject.

infoboxes - parsed information from the side panel (infobox) on the Wikipedia article.

sections - parsed sections of the article, including links.

Note: excludes other media/images, lists, tables and references or similar non-prose sections.



Kaggle dataset: People

Data Structure

File name: wme_people_infobox.tar.gz

Size of compressed file: 4.12 GB

Size of uncompressed file: 21.28 GB

Noteworthy Included Fields:

name - title of the article.

identifier - ID of the article.

image - main image representing the article's subject.

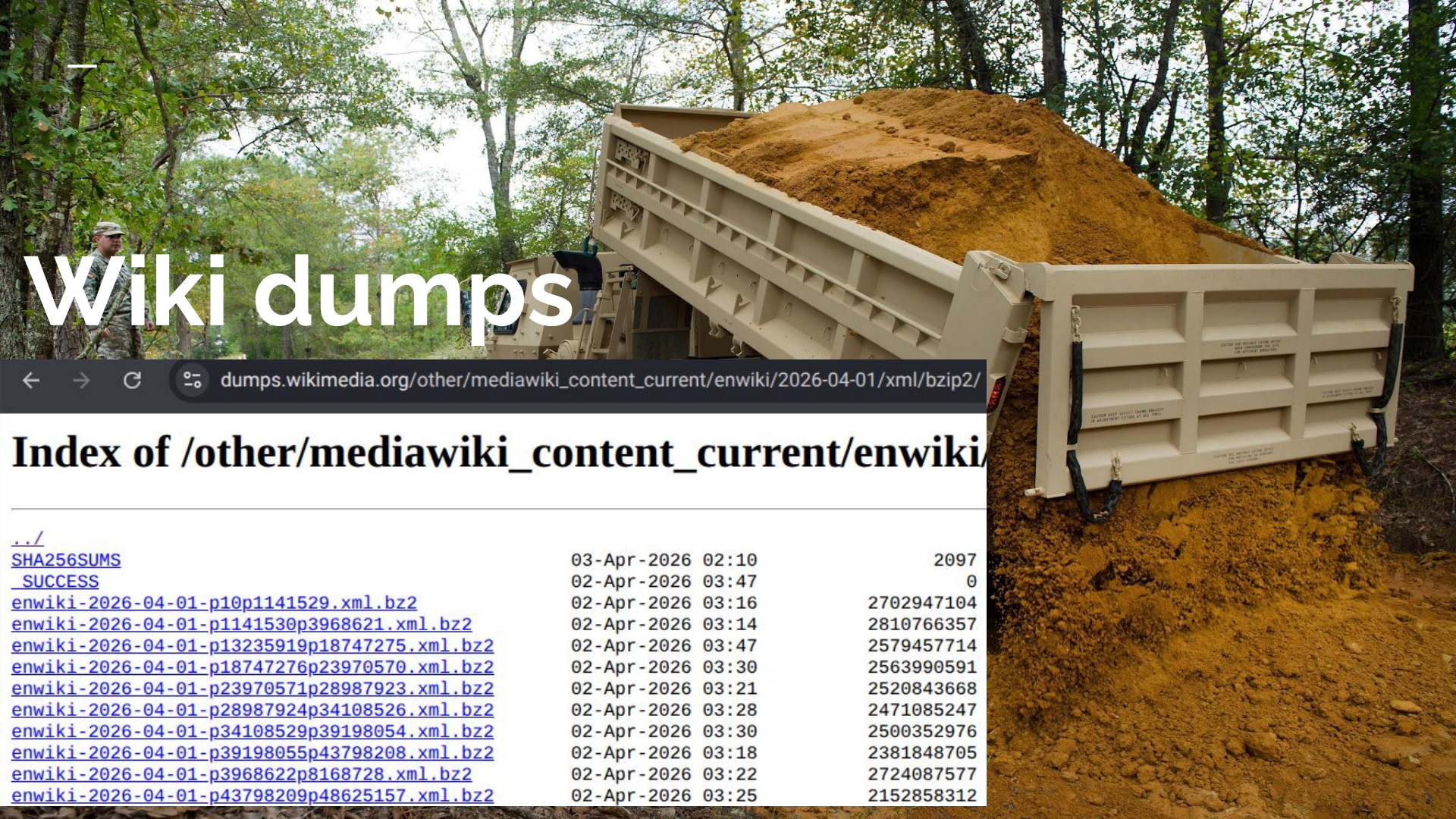
description - one-sentence description of the article for quick reference.

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Note: excludes other media/images, lists, tables and references or similar non-prose sections.



Wiki dumps

← → ↺ 🌐 dumps.wikimedia.org/other/mediawiki_content_current/enwiki/2026-04-01/xml/bzip2/

Index of /other/mediawiki_content_current/enwiki/

../		
SHA256SUMS	03-Apr-2026 02:10	2097
SUCCESS	02-Apr-2026 03:47	0
enwiki-2026-04-01-p10p1141529.xml.bz2	02-Apr-2026 03:16	2702947104
enwiki-2026-04-01-p1141530p3968621.xml.bz2	02-Apr-2026 03:14	2810766357
enwiki-2026-04-01-p13235919p18747275.xml.bz2	02-Apr-2026 03:47	2579457714
enwiki-2026-04-01-p18747276p23970570.xml.bz2	02-Apr-2026 03:30	2563990591
enwiki-2026-04-01-p23970571p28987923.xml.bz2	02-Apr-2026 03:21	2520843668
enwiki-2026-04-01-p28987924p34108526.xml.bz2	02-Apr-2026 03:28	2471085247
enwiki-2026-04-01-p34108529p39198054.xml.bz2	02-Apr-2026 03:30	2500352976
enwiki-2026-04-01-p39198055p43798208.xml.bz2	02-Apr-2026 03:18	2381848705
enwiki-2026-04-01-p3968622p8168728.xml.bz2	02-Apr-2026 03:22	2724087577
enwiki-2026-04-01-p43798209p48625157.xml.bz2	02-Apr-2026 03:25	2152858312

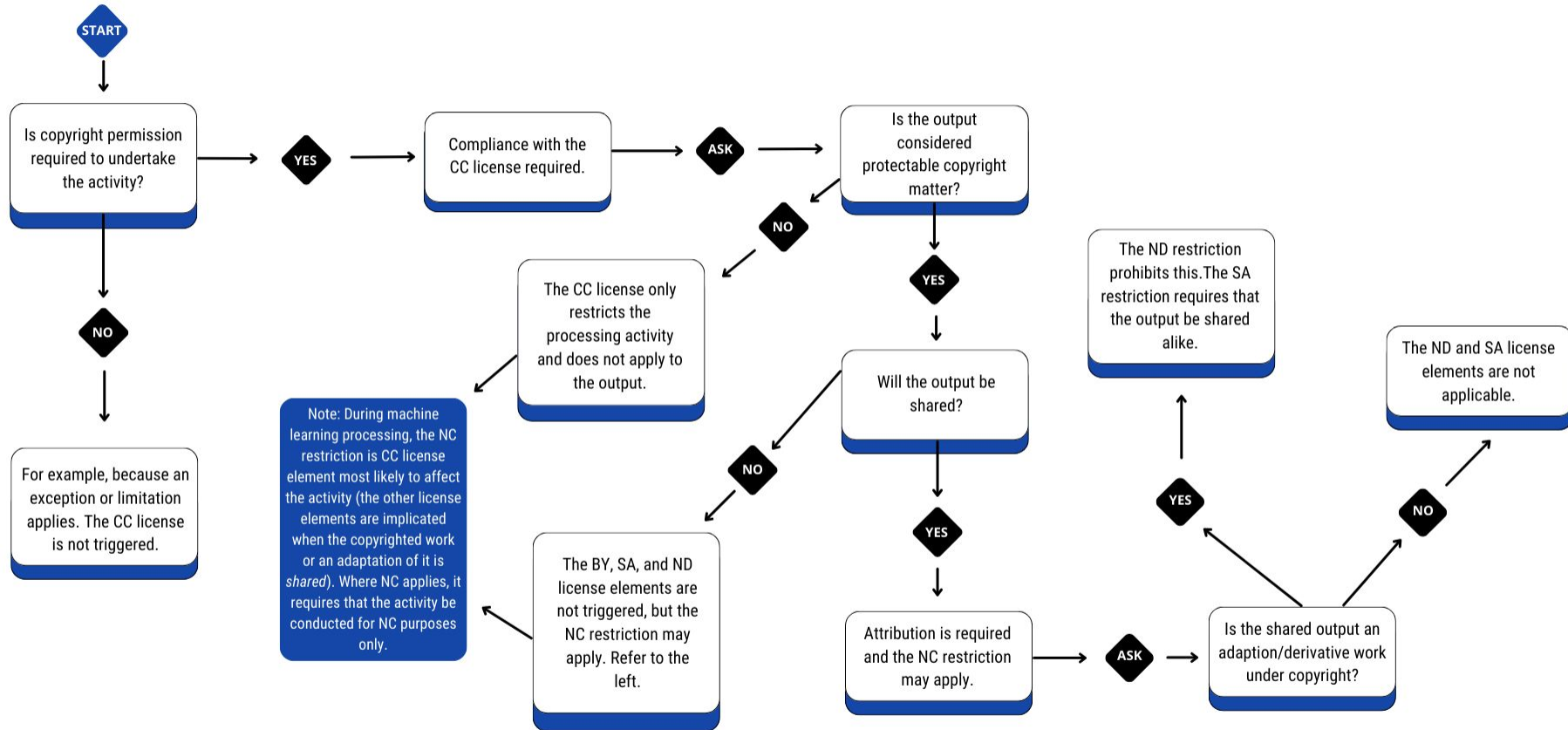
The Weaponisation of openness

Stefaan G. Verhulst (Oct 2025)

open resources have been mined at planetary scale for AI training. This process has largely taken place without **meaningful consent**, compensation, or even **attribution**. The result is a modern-day twist on the tragedy of the commons: The more open the digital ecosystem became, the more it was extracted.

The considerate scraper's code of conduct

- Minimise impact (server load, emissions) of repeated scraping
 - Reuse in-memory copy of scraped data
 - Put scraped data in local/cloud storage or database
- Be aware of Acceptable Use Policies, Fair Use etc.
- Respect the licence/copyright, if applicable.
 - Allow for lineage/attribution
- Work with the data provider, not against them
 - Use provided APIs, wrapper libraries
 - Use alternative sources intended for querying e.g. Wikidata



UK exceptions to copyright

Text and data mining for non-commercial research

An exception to copyright exists which allows researchers to make copies of any copyright material for the purpose of computational analysis if they already have the right to read the work (that is, they have 'lawful access' to the work). This exception only permits the making of copies for the purpose of text and data mining for non-commercial research. Researchers will still have to buy subscriptions to access material; this could be from many sources including academic publishers.

<https://www.gov.uk/guidance/exceptions-to-copyright>

Thank you:

DataLab events: Elena Simperl, Jimmy Wales
Python Glasgow meetup
Green Software Foundation meetup
Open Data Camp Edinburgh 2025

Stelios Christodoulou
github.com/stelios-c

